

Machine Learning and Artificial Intelligence Applied to Computational Arts, Music, and Games

Kıvanç Tatar, Assistant Professor in Interactive AI



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Overview

- Background
 - AI in Creative Domains
 - Why does it matter?
- Research Achievement: Bridging STEM with Social Sciences, Humanities and Arts
- Foundation of the Research Group
- Future Research directions

These slides are live at:

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Background: Arts and Music in the Era of Artificial Intelligence

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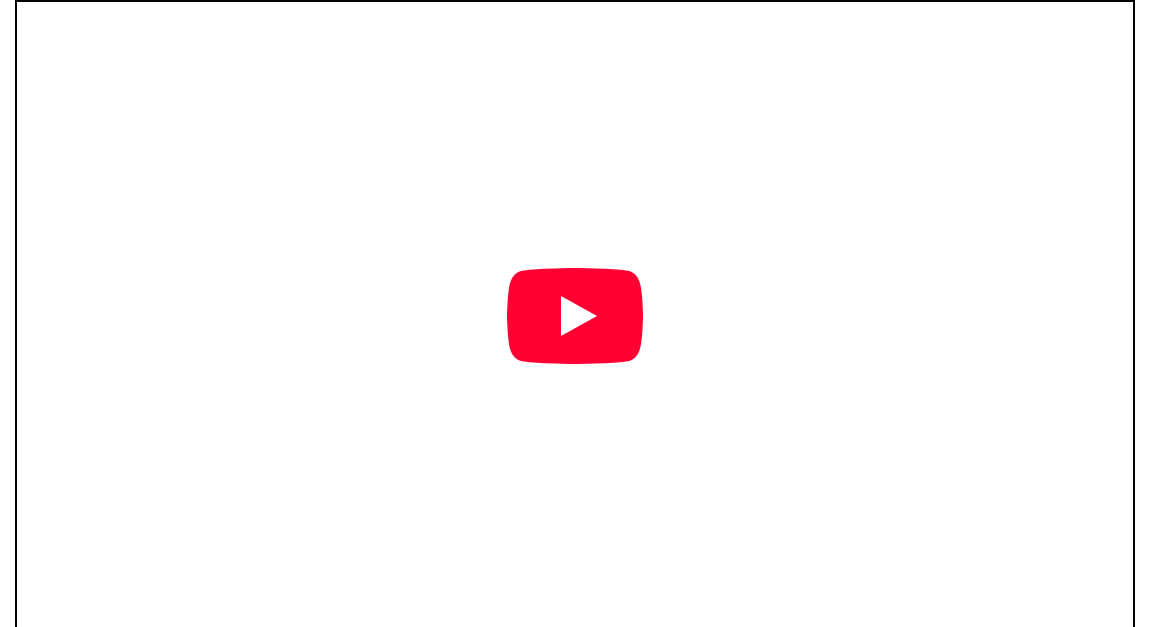
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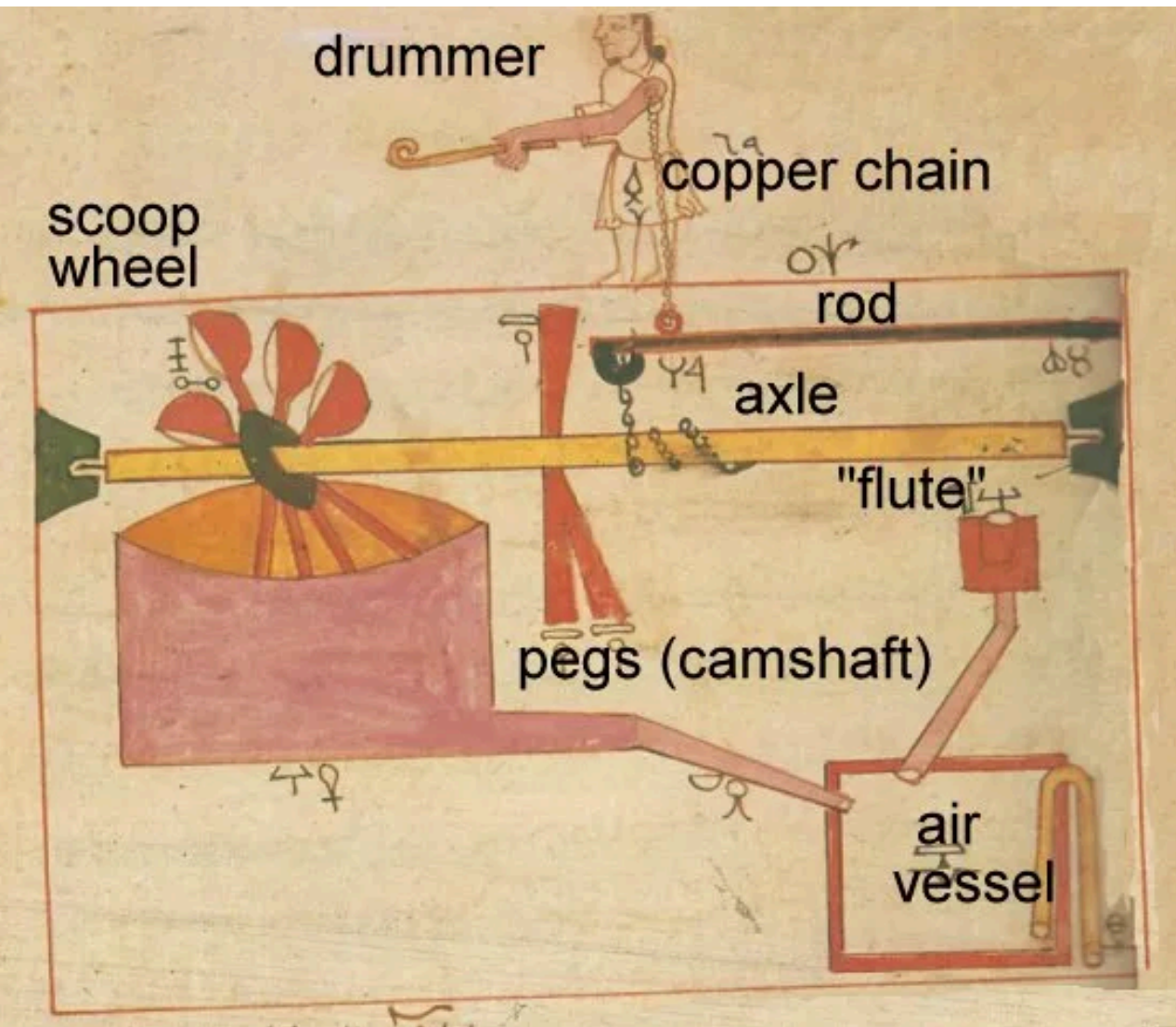


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Creative Coding,

- is a type of computer programming activity where the end result is a creative outcome.
- does not necessarily focus on solving a problem or functionality.
- is an exploration of aesthetics, experiences, forms, creative affordances and agency.





Before ML and AI: Generative Arts

- Autonomy and Computational Thinking
- Agency and Interaction

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Arts and Music in the Era of Artificial Intelligence

- Highly autonomous tools
- Increasingly complex interaction



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Why does it matter?

- Arts and Music are core components of daily life
- A major shift in all multimedia domains due to AI
- The global multimedia industry such as music, games, movies etc. is one of the largest:
 - Music: "Estimated at \$105 billion in 2024, with expectations to nearly double to \$200 billion by 2035" [1],
 - Video Games: "A \$189 billion market in 2025" [2],
 - Video Streaming: "In 2024, ...reached over 316 billion U.S. dollars" [3].

[1] Goldman Sachs. Global Music Revenues Are Forecast to Double to \$200 Billion in 2035. (2025). Retrieved January 1, 2026, from

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Research Achievement: Bridging STEM with Social Sciences, Humanities and Arts

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Research Achievements so far (STEM)

Human-AI interactions in artistic contexts

- How to utilize vast AI search spaces in content generation (ie sound, music, image etc.) (J10, J9, C14)
- Focused on process of AI technology development rather than AI generated products (J8, C18)
- Applied research through design processes to AI approaches (C18)

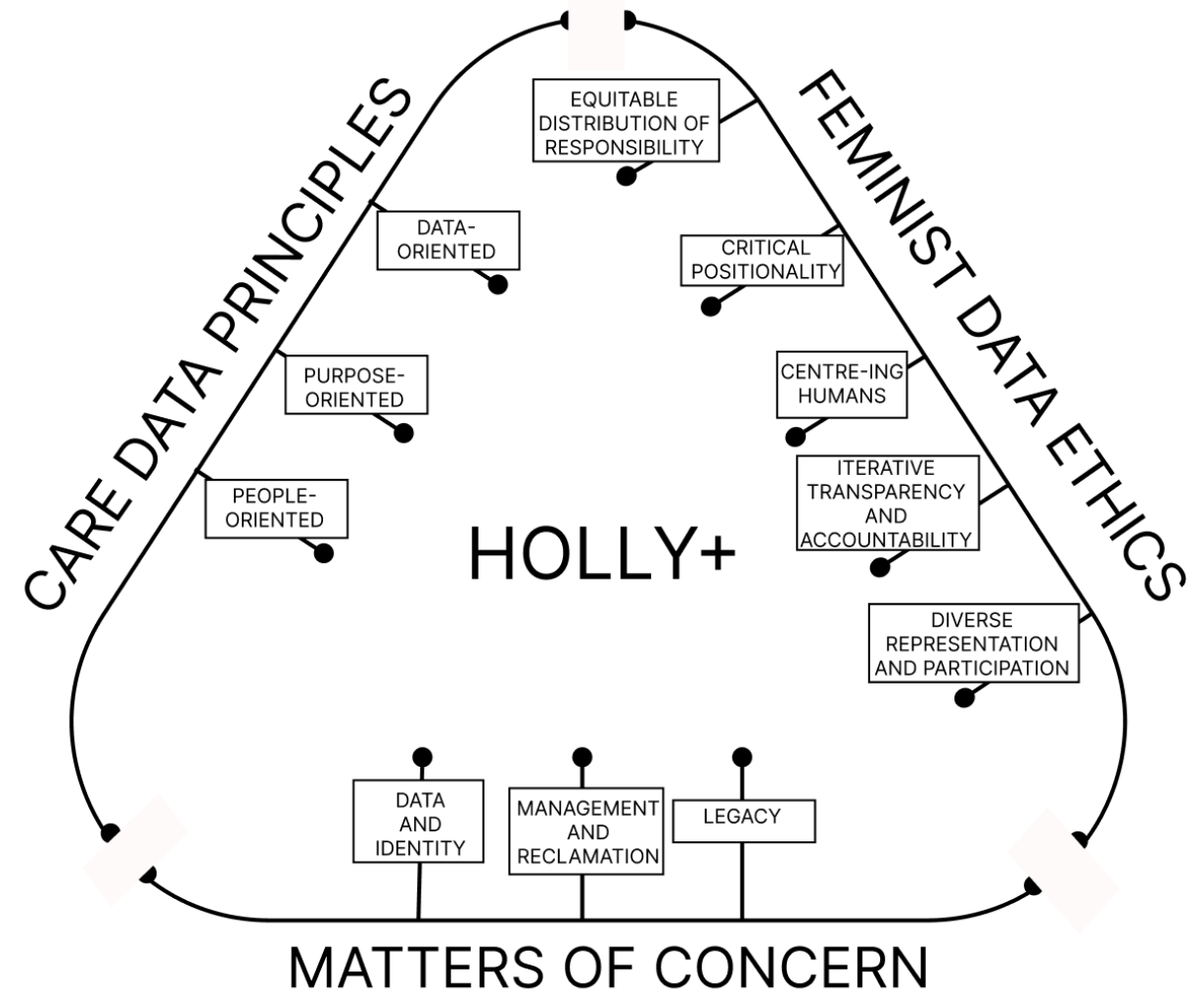
Research Achievements so far (STEM)

Founded new directions in future Musical AI research

- Neural Audio Instruments (J9)
- Musical Acoustics utilizing Deep Learning (C19)
- Creativity and Games (J12)

Research Achievements so far (SSH)

- Interdisciplinary research methods
 - Bridging technology facts to societal matters (J8, J7, C16, C15)
- AI Ethics towards informing future legislations (J13, J7, C16, C15, J11)
 - Voice AI and future legislations (J13, C16)
- Future of artistic practices in the era of AI (J7, J11)



Major Achievement: The Foundation of the Research Group

AICompArts : AI in Computational Arts, Music, and Games

- Bridging STEM with Humanities and Arts
- Strong connection to research and teaching strategies of Chalmers (TRA385)
- Leading national and international research domain of Musical AI and AI arts
- AI innovation with societal considerations at the center

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Research Vision

- Computational autonomy informed by theory of the application domain (in our case music and arts)
- Societal implications of ML and AI as one of the core aspects of technology making processes
- Artistic curiosity as a driving force for research and innovation

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AI in Computational Arts, Music, and Games

AI in Computational Arts, Music and Games research group focuses on exploring the novel aesthetics and affordances of Machine Learning (ML) and Artificial Intelligence (AI) within artistic applications.

We conduct research into+with+through algorithmic approaches such as deep generative models, supervised, unsupervised, and reinforcement learning, evolutionary approaches, multi-agent systems, and natural language processing. The artistic applications include musical and audiovisual performances, interactive installations, game art, augmented and virtual reality, video art, art with robotics, and bio-art. Artistic concepts and specific applications guide our technological and algorithmic choices, rather than independent disciplinary silos. We position this interdisciplinary approach under an encompassing term, Computational Arts.

Conventional Machine Learning and Artificial Intelligence emphasize gestalt—which is the end-product—and optimization while addressing questions such as: which architecture performs better or faster? How do we generate high-quality content such as image/video/audio/music? While these questions and gestalt are relevant with their significant contributions to society as the matters of fact, investigations into the AI technology creation processes, actors, and stakeholders can benefit societal discourses towards matters of concerns. Artistic realms provide opportunities to investigate AI technology both as their technology facts of new artistic opportunities in greater autonomy, and as their societal and cultural matters of concern.

The perspectives from third-wave feminism combined with methodologies from Science Technology Studies and philosophy of technology can create bridges that connect matters of facts such as accuracy, efficiency, optimization, reliability, and explainability; to matters of concerns such as ethics, inclusivity, sustainability, accessibility, and privacy. A better future with technologies of greater autonomy, can only be achieved with interdisciplinary perspectives. With artistic imagination in technology-making, informed by care-fully troubling processes, we pursue novel interdisciplinary research in artistic applications of Machine Learning and AI.

The group is led by Kıvanç Tatar. More information about the research group can be found on the [external research group page](#).

Members



[Kivanc Tatar](#)

Doctor, Data Science and AI, Computer Science and Engineering



[Kelsey Cotton](#)

Doctoral Student, Data Science and AI, Computer Science and Engineering



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Research Questions for future directions

- How do we inform AI technology development with considerations in societal implications?
- What are new artistic affordances of novel AI architectures?
- How can we bring back to "the body" to digital AI architectures?

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Specifically, what are future directions?

- Proposing mixed research methods to **analyze cultural dimensions encoded within AI** models
 - Quantitative metrics to analyze entanglements between culturally relevant dimensions
 - Qualitative methods to critically engage with the findings of quantitative metrics
- Leading the future **AI technology in creative domains**:
 - Neural Audio Instruments: a new category of musical instruments
 - Neural Acoustics: towards real-time acoustics simulations for sound synthesis
 - Co-creation of video games using AI generated code
- Anything but text!**
 - Utilizing movement computing in human-AI interaction

Why are these directions important?

- Proposing mixed research methods to analyze cultural dimensions encoded within AI models.
A limited number of AI models are globally deployed. Their biases (currently unknown) have been shifting all aspects of society.
- Neural Audio Instruments and Neural Acoustics:
It is important to show good practices in using ML and AI in artistic contexts, addressing the current uprisings against AI in artistic communities
- Anything but text!*
Current AI development is overtly focused on text-to-X models, yet there is more to being a human than text. We have a body and we can move in 3D!

Over-arching Goal - Bridging AI Technology and Engineering Research with Humanities and Social Sciences

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Research Community Building

- Local community building
 - Culture Code event in February
 - Future plans for events with Universium etc.
- National research community building for international impact: WASP-HS cluster and the VR Research Environment in SSH
- Leading musical AI and AI arts research fields internationally

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Seminar ● Open for registration

The Culture Code: Generative AI, Data, and Societal Futures

This event, part of the AI Futures of Culture and Memory WASP-HS research cluster and funded by the VR networking project The Culture Code: Developing Deep Sustainable AI, brings together scholars, practitioners, and artists to examine the societal impact of deep generative models.

Overview

● Open for registration

Date:

Starts 13 February 2026, 08:45
Ends 13 February 2026, 18:00

Location:

📍 A Working Lab Studio, Sven Hultins plats 5, Campus Johanneberg

🖥 [Event page](#)

Language:

English

Last sign up date:

4 February 2026

[Register here](#)

Generative AI technologies are transforming cultural production, creative practices, and societal narratives at an unprecedented scale. While these systems offer new opportunities for democratizing access to cultural resources and enabling innovative forms of expression, they also raise critical concerns around bias, representation, sustainability, and ethics. This event, part of the AI Futures of Culture and Memory WASP-HS research cluster and funded by the VR networking project The Culture Code: Developing Deep Sustainable AI, brings together scholars, practitioners, and artists to examine the societal impact of deep generative models.

February
13

08:45 - 18:00

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Current Funding

WASPHS-3: WASP-HS Research Cluster Call. AI Futures of Culture and Memory. 30 million SEK.

VR-1: Vetenskapsrådet, Research environment grant - humanities and social sciences. VOICE. AI-generated voices. Legal and societal perspectives. 18 million SEK.

WASPHS-1: Interactive AI - Ethics and Aesthetics of Human-Machine Interaction in Art, Music, and Games. 12 million SEK

Submitted Projects Awaiting Decision

ERC-1 European Research Council, Starting Grant. Multimodal Generative Modelling for Interpretable Movement-to-Audio Synthesis: MODA. Kivanç Tatar (PI, Chalmers University of Technology and University of Gothenburg). 1.5 million Euro ≈16.4 million SEK.

Thank you!

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